

TANGUY PEMEJA

Senior Backend & Platform Engineer

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📍 Japan - Relocating to Singapore
(Dependant's Pass)

SUMMARY

Senior Backend & Platform Engineer with 5 years of experience building backend APIs and distributed systems across cloud and edge environments in cybersecurity and IoT domains. Strong focus on FastAPI-based backend development, platform engineering, and production operations, combined with experience in infrastructure automation, deployment workflows, and observability.

Built and maintained backend services for device lifecycle management, configuration delivery, and system orchestration, alongside internal platforms supporting deployment and monitoring of production systems.

Currently working on AI-enabled edge computing systems for device management, developing APIs for device lifecycle control, managing containerized runtimes, and implementing observability across distributed Linux-based infrastructure. Experienced working in international environments across France and Japan, collaborating with teams across multiple time zones.

WORK EXPERIENCE

Senior Backend Developer

EDGEMATRIX Inc • Tokyo, Japan

May 2024 – Present

- Designed and maintained backend APIs and cloud services for an IoT edge platform supporting device lifecycle management, configuration delivery, and synchronization with Linux-based edge devices, including a staging environment for deployment validation.
- Implemented cloud-side authentication and authorization using Auth0 (OAuth2/OIDC), securing API access and managing user and service permissions within the cloud platform.
- Contributed to a secure update and deployment pipeline for edge devices, supporting encrypted, device-specific software distribution and improving build and delivery workflows.
- Maintained a Docker Compose-based microservices runtime on edge devices (~10 services per device), managing service composition, routing, and service interactions using Traefik and Authelia.
- Designed and operated monitoring systems across cloud and edge environments, using AWS CloudWatch for cloud-side visibility and Grafana + Loki for edge observability and log analysis.

Cybersecurity Engineer

TEHTRIS Japan • Tokyo, Japan

Dec 2022 – Apr 2024

- Developed an internal self-service deployment platform enabling automated provisioning of infrastructure and custom OS environments on OVH Cloud, leveraging Airflow and Packer to reduce deployment time from multiple days to a few hours.
- Served as primary incident responder during APAC coverage hours, investigating production anomalies through logs and metrics, coordinating escalations, and contributing to post-incident reviews with global engineering teams.
- Designed and evolved Grafana monitoring dashboards for production systems, introducing new metrics, visualizations, and monitoring thresholds to improve system visibility and accelerate issue detection.
- Supported technical investigations during APAC coverage hours for XDR-related products, reproducing issues, gathering diagnostic information, and coordinating with engineering teams to accelerate resolution of complex cases.

- Developed and maintained FastAPI-based backend services exposing APIs for external XDR security products, enabling customers and external tools to interact with XDR platform services.
- Built an internal operations platform (backend APIs + lightweight HTML/CSS web interface) to manage OVH Cloud environments and operational workflows, implementing authentication and role-based access control using Okta.
- Improved reliability of owned services through API hardening, testing, and documentation automation, ensuring safe and controlled exposure of critical security and operational endpoints for external usage.

- Developed and optimized neural network-based control systems for a prosthetic arm, improving movement precision through simulation and experimental analysis.
- Implemented a virtual reality simulation environment for data collection, testing, and evaluation of prosthetic arm control models.

EDUCATION

Diplôme d'Ingénieur (MSc Computer Science)

ENSEIRB-MATMECA | 2018–2021

Exchange Semester

Keio University | 2020 – 2021

LANGUAGE

French Native

English Full Professional

Japanese Elementary

TECHNICAL SKILLS

Backend

Python, FastAPI, Pydantic, WebSockets, REST APIs, OpenAPI/Swagger

Frontend

Vue.js, PrimeVue, Tailwind CSS, HTML, JavaScript

Cloud & Platform Engineering

AWS (EC2, Elastic Beanstalk, CloudWatch), OVH Cloud, Docker, CI/CD, GitHub Actions, Airflow, Packer

Observability & Operations

Grafana, Loki, Prometheus, Monitoring, Alerting, Incident Response

Edge & Systems

Linux, Docker Compose, Traefik, IoT / Edge Device Management

Databases

MySQL, Redis

Identity & Access Management

OAuth2/OIDC, RBAC, Auth0, Okta, Authelia

Security

XDR (EDR, SIEM, SOAR)